

Novel NDVI-Based Vegetation Stress Analysis for Forest Fire Susceptibility Mapping in India

Methodology Section — Research Working Document

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1 NDVI-Based Vegetation Stress Analysis for Forest Fire Susceptibility

1.1 Overview and Motivation

The Normalized Difference Vegetation Index (NDVI) is defined as:

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + \rho_{\text{Red}}} \quad (1)$$

where ρ_{NIR} and ρ_{Red} denote the surface reflectance in the Near-Infrared and Red spectral bands respectively. The index ranges from -0.2 to 1.0 , with higher values indicating denser and healthier vegetation.

NDVI emerged as the **single most important predictor** of forest fire occurrence in India among 15 environmental variables, contributing 22.3% permutation importance and 28.4% percentage contribution in the MaxEnt framework of ?. Despite this primacy, prior studies — including ? — employed NDVI solely as a static or temporally-averaged predictor, foregoing the rich temporal information encoded in the 22-year MODIS satellite record (2000–2022).

Methodological Gap Addressed in This Study

? applied NDVI as:

- A single time-averaged raster layer
- Without pixel-level quality filtering (cloud/snow contamination retained)
- Without temporal decomposition (seasonal cycle confounds prediction)
- Without pre-fire stress signal extraction
- Assuming a linear NDVI–fire relationship

This study introduces five mathematically grounded extensions that transform raw NDVI into physically interpretable, temporally dynamic, and spatially aware predictors of fire susceptibility.

The five novel components introduced in this study are:

1. **QA-filtered scale-corrected NDVI** (Section 1.2)
2. **Seasonal decomposition and anomaly extraction** (Section 1.3)
3. **Pre-fire Cumulative Vegetation Stress Index (CVSI)** (Section 1.4)
4. **Spatial autocorrelation and fire-cluster alignment** via Moran’s I (Section 1.5)
5. **NDVI–fire threshold detection via breakpoint analysis** (Section 1.6)

1.2 Data Source and Preprocessing

NDVI data were obtained from the MODIS/Terra Vegetation Indices Monthly product **MOD13C2 Version 6.1** (?), hosted at the NASA Land Processes Distributed Active Archive Center (LP DAAC) and accessed via the AppEEARS extraction service (?). The product provides monthly composite NDVI at 0.05° (≈ 5.6 km) spatial resolution in the Climate Modeling Grid (CMG) projection. A higher-resolution variant (MOD13A3 v061, 1 km monthly) was also evaluated and yields improved spatial detail over the 0.05° product used in ?. The study period spans February 2000 through August 2022 ($T = 272$ monthly composite images).

1.2.1 Scale Correction and Fill-Value Masking

Raw MOD13C2 stores NDVI as signed 16-bit integers. The corrected floating-point NDVI for pixel (i, j) at time step t is:

$$\text{NDVI}_{ij}^{(t)} = \begin{cases} \text{raw}_{ij}^{(t)} \times 10^{-4} & \text{if } -2000 \leq \text{raw}_{ij}^{(t)} \leq 10000 \\ \text{NaN} & \text{otherwise} \end{cases} \quad (2)$$

yielding true $\text{NDVI} \in [-0.2, 1.0]$. The designated fill value of -3000 and all values outside the valid range are replaced with NaN prior to any analysis.

1.2.2 Quality Assurance Filtering

Novel Contribution 1: Pixel-Level QA Filtering

? applied no pixel-level quality screening to NDVI, implicitly including cloud-contaminated and snow/ice-affected retrievals. This work applies the `pixel_reliability` Science Data Set (SDS) from MOD13C2 to filter each pixel before analysis.

The QA-filtered NDVI is defined as:

$$\text{NDVI}_{ij}^{(t), \text{QA}} = \begin{cases} \text{NDVI}_{ij}^{(t)} & \text{if } q_{ij}^{(t)} \in \{0, 1\} \\ \text{NaN} & \text{otherwise} \end{cases} \quad (3)$$

where pixel reliability values are defined as: $q = 0$ (Good data, use without further screening), $q = 1$ (Marginal data, useful but inspect detailed QA), $q = 2$ (Snow/Ice — exclude), $q = 3$ (Cloudy — exclude), and $q = -1$ (Fill/No data) (?). QA filtering is especially critical for monsoon months (June–September), when cloud contamination

over India is highest and unfiltered NDVI values are systematically negatively biased.

1.3 Seasonal Decomposition and Anomaly Extraction

1.3.1 Climatological Mean and Monthly Anomaly

India's vegetation phenology is dominated by a strong annual monsoon cycle. To isolate the inter-annual fire-relevant signal from the seasonal background, the long-term climatological mean for calendar month m at pixel (i, j) is computed over the 2001–2020 baseline (consistent with ?):

$$\bar{\mu}_{ij}^{(m)} = \frac{1}{|\mathcal{Y}_m|} \sum_{y \in \mathcal{Y}_m} \text{NDVI}_{ij}^{(y,m), \text{QA}} \quad (4)$$

where $\mathcal{Y}_m = \{2001, 2002, \dots, 2020\}$ and only non-NaN values contribute. The **monthly NDVI anomaly** at year y , month m , pixel (i, j) is:

$$\delta_{ij}^{(y,m)} = \text{NDVI}_{ij}^{(y,m), \text{QA}} - \bar{\mu}_{ij}^{(m)} \quad (5)$$

Negative anomaly ($\delta < 0$) indicates below-average vegetation density (stress); positive ($\delta > 0$) indicates above-average greenness. This operation removes the confounding seasonal cycle and exposes the inter-annual vegetation stress signal directly relevant to fire ignition.

1.3.2 STL Decomposition for Trend Isolation

Novel Contribution 2: STL Temporal Decomposition

Biswas et al. (?) did not perform any temporal decomposition of NDVI. This study decomposes the 22-year pixel-wise time series using STL (Seasonal and Trend decomposition using LOESS), separating long-term vegetation trend from the seasonal and residual components.

The STL decomposition (?) is applied to each spatial pixel's complete time series:

$$\text{NDVI}_{ij}^{(t)} = T_{ij}^{(t)} + S_{ij}^{(t)} + R_{ij}^{(t)} \quad (6)$$

where:

- $T_{ij}^{(t)}$ — **Trend component**: captures inter-annual vegetation change driven by deforestation, afforestation, and land-use change. A declining trend ($\partial T / \partial t < 0$)

over a forest pixel indicates vegetation loss and elevated fire risk.

- $S_{ij}^{(t)}$ — **Seasonal component**: the repeating annual monsoon-driven phenology cycle (period = 12 months).
- $R_{ij}^{(t)}$ — **Residual component**: captures anomalous events including drought episodes, pest outbreaks, pre-fire fuel drying, and post-fire regeneration.

The **Mann–Kendall trend test** (??) is applied pixel-wise to $T_{ij}^{(t)}$ to identify statistically significant greening or browning trends at the $\alpha = 0.05$ significance level. The Mann–Kendall statistic τ is:

$$\tau = \frac{P - Q}{\frac{1}{2}n(n-1)} \quad (7)$$

where P is the number of concordant pairs and Q the number of discordant pairs in the time series. Pixels with $\tau < 0$ and $p < 0.05$ are classified as **significant browning** zones — regions of long-term vegetation decline contributing to elevated fire susceptibility.

1.4 Cumulative Vegetation Stress Index (CVSI)

Novel Contribution 3: Pre-Fire Vegetation Stress Quantification

No prior India-scale forest fire susceptibility study has introduced a temporally-accumulating pre-fire NDVI stress metric. The CVSI defined below provides a direct proxy for fuel moisture deficit in the weeks to months preceding ignition, grounded in combustion physics.

1.4.1 Mathematical Formulation

Let τ_f denote the month of fire occurrence at pixel (i, j) . The k -month **Cumulative Vegetation Stress Index (CVSI)** leading up to τ_f is:

$$\text{CVSI}_{ij}(\tau_f, k) = \sum_{t=\tau_f-k}^{\tau_f-1} \max\left(-\delta_{ij}^{(t)}, 0\right) \quad (8)$$

The operator $\max(-\delta, 0)$ retains only negative anomaly episodes (stress) and discards above-average greenness months. For $k = 3$:

$$\begin{aligned}
\text{CVSI}_{ij}(\tau_f, 3) = & \max\left(-\delta_{ij}^{(\tau_f-3)}, 0\right) \\
& + \max\left(-\delta_{ij}^{(\tau_f-2)}, 0\right) \\
& + \max\left(-\delta_{ij}^{(\tau_f-1)}, 0\right)
\end{aligned} \tag{9}$$

Physical justification: Prolonged below-average NDVI signals reduced canopy moisture content, accelerated fine fuel drying, and increased bulk density of dead surface fuels susceptible to ignition (?). The CVSI thus quantifies the *pre-fire fuel moisture deficit*, a well-established physical precursor to wildfire ignition (?).

1.4.2 Optimal Lag Selection via Mutual Information

The optimal accumulation window k^* is selected by maximising the **mutual information** (MI) between $\text{CVSI}(\tau_f, k)$ and binary fire occurrence labels $F_{ij} \in \{0, 1\}$:

$$k^* = \arg \max_{k \in \{1,2,3,4,5,6\}} I(\text{CVSI}(\tau_f, k) ; F_{ij}) \tag{10}$$

where the Shannon mutual information is:

$$I(X; Y) = \sum_{x,y} p(x, y) \log_2 \frac{p(x, y)}{p(x) p(y)} \tag{11}$$

1.4.3 Comparison with Existing Fire Danger Indices

Table 1 compares the CVSI with commonly used fire danger indices.

Table 1: Comparison of CVSI with existing vegetation-based fire indices

Index	Input	Temporal depth	Available for India?
Raw NDVI (?)	NDVI	Single month	Yes
NDWI	NIR, SWIR	Single month	Yes
FWI (Canadian)	Climate variables	Daily rolling	Limited
KBDI	Precipitation, T	Monthly	Limited
CVSI (this work)	NDVI anomaly	k^* months pre-fire	Yes (national)

1.5 Spatial Autocorrelation: Global and Local Moran's I

Novel Contribution 4: Spatial Neighbourhood of Vegetation

? treated all spatial pixels as independent observations. This work explicitly models the spatial dependency structure of NDVI using Moran's I , capturing the fire-relevant information embedded in a pixel's vegetation neighbourhood.

1.5.1 Global Moran's I

For a spatial field $\{x_{ij}\}$ with N pixels and row-standardised spatial weight matrix $\mathbf{W}$ (Queen contiguity):

$$I = \frac{N}{\sum_{i,j} \sum_{k,l} w_{ij,kl}} \cdot \frac{\sum_{i,j} \sum_{k,l} w_{ij,kl} (x_{ij} - \bar{x})(x_{kl} - \bar{x})}{\sum_{i,j} (x_{ij} - \bar{x})^2} \quad (12)$$

Under spatial randomness, $\mathbb{E}[I] = -1/(N-1)$. Statistical significance is assessed via permutation testing ($n_{\text{perm}} = 999$) at $\alpha = 0.05$. $I > 0$ indicates spatial clustering; $I < 0$ indicates dispersion.

1.5.2 Local Indicators of Spatial Association (LISA)

The Local Moran's I statistic at pixel (i, j) is:

$$I_{ij} = \frac{x_{ij} - \bar{x}}{S^2} \sum_{k,l} w_{ij,kl} (x_{kl} - \bar{x}) \quad (13)$$

where $S^2 = \frac{1}{N} \sum_{i,j} (x_{ij} - \bar{x})^2$. LISA classification yields four spatial cluster types:

Table 2: LISA quadrant classification and fire-risk interpretation

LISA Class	Description	Fire Risk Implication
High–High (HH)	High NDVI, high-NDVI neighbourhood	Dense contiguous forest; interior fire risk
Low–Low (LL)	Low NDVI, low-NDVI neighbourhood	Degraded/sparse forest; low fuel load
High–Low (HL)	High NDVI, low-NDVI neighbourhood	Forest fragment at WUI; elevated edge fire
Low–High (LH)	Low NDVI, high-NDVI neighbourhood	Recent burn scar / clearcut within forest

The LISA cluster class is incorporated as a **categorical feature** in the ML pipeline, encoding spatial neighbourhood context that pixel-level NDVI values alone cannot express.

1.6 NDVI–Fire Threshold Detection via Breakpoint Analysis

Novel Contribution 5: Non-linear NDVI–Fire Relationship

? implicitly assumed a monotonic linear relationship between NDVI and fire probability within the MaxEnt framework. This study explicitly models the **non-linearity** of the NDVI–fire relationship by detecting the critical threshold θ^* below which fire occurrence probability increases sharply.

1.6.1 Piecewise Logistic Fire–NDVI Model

Let $P(F = 1 \mid \text{NDVI} = x)$ denote the probability of fire occurrence given pixel NDVI value x . A piecewise logistic model with a single structural breakpoint θ is:

$$P(F = 1 \mid x) = \begin{cases} \sigma(\alpha_1 + \beta_1 x) & \text{if } x \leq \theta \\ \sigma(\alpha_2 + \beta_2 x) & \text{if } x > \theta \end{cases} \quad (14)$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the logistic sigmoid. The parameter vector $\boldsymbol{\psi} = \{\alpha_1, \beta_1, \alpha_2, \beta_2, \theta\}$ is estimated by minimising the binary cross-entropy (negative log-likelihood):

$$\hat{\boldsymbol{\psi}} = \arg \min_{\boldsymbol{\psi}} \mathcal{L}(\boldsymbol{\psi}) = - \sum_{n=1}^N \left[F_n \log P_n + (1 - F_n) \log(1 - P_n) \right] \quad (15)$$

The breakpoint θ^* is found via grid search over the 10th–90th percentile of observed NDVI followed by gradient-based fine-tuning (Nelder–Mead optimisation). The estimated θ^* represents the **critical vegetation density threshold for fire ignition** in each biogeographic zone.

1.6.2 Biogeographic Zone-Stratified Thresholds

India’s forest types span five major biogeographic zones (?): (1) Northeastern tropical, (2) Western Ghats tropical wet, (3) Deccan dry deciduous, (4) Central Indian mixed, and (5) Himalayan sub-alpine. A single national θ^* is ecologically inadequate. Zone-stratified thresholds $\{\theta_z^*\}_{z=1}^5$ are estimated independently and incorporated as zone-specific NDVI risk indicators in the final susceptibility mapping.

1.7 NDVI Feature Engineering Summary

Table 3 summarises all NDVI-derived features entering the ML pipeline, expanding from the single raw NDVI layer of ? to a nine-feature NDVI representation.

Table 3: Complete NDVI-derived feature set for forest fire susceptibility mapping

ID	Feature	Equation	Physical Meaning	Novel?
F1	QA-filtered NDVI $\overline{\text{NDVI}}_{\text{QA}}$	Eq. (3)	Mean vegetation density (cloud-screened)	Partial
F2	Climatological mean $\bar{\mu}^{(m)}$	Eq. (4)	Long-term seasonal baseline	No
F3	Monthly anomaly δ	Eq. (5)	Inter-annual vegetation stress	Yes
F4	STL trend T	Eq. (6)	22-year vegetation change trajectory	Yes
F5	STL residual R	Eq. (6)	Anomalous events (drought, fire scar)	Yes
F6	Mann–Kendall τ	Eq. (7)	Browning/greening trend significance	Yes
F7	CVSI(k^*)	Eq. (8)	Pre-fire fuel moisture deficit	Yes
F8	Local Moran’s I_{ij} / LISA	Eq. (13)	Spatial neighbourhood vegetation cluster	Yes
F9	$\nVdash[\text{NDVI} < \theta_z^*]$	Eq. (14)	Critical ignition threshold indicator	Yes
<i>Biswas et al. (2025): 1 NDVI feature</i>				—
<i>This work: 9 NDVI-derived features</i>				7 novel

1.8 Integration with MaxEnt and ML Models

The nine NDVI-derived features (Table 3) are integrated into the prediction pipeline alongside the 14 other variables of ? (Table 2 therein), yielding a **23-dimensional feature space** compared to the 15-dimensional space of the baseline study. In the MaxEnt framework, each NDVI feature enters as a separate continuous environmental layer. In the supervised ML pipeline (Random Forest, XGBoost, SVM), SHAP (SHapley Additive exPlanations) analysis (?) is applied to decompose each feature’s marginal contribution — providing model-agnostic attribution that extends beyond the permutation importance reported in ?.